MILITARY SPECIFICATION

SHIPPING AND STORAGE CONTAINER, CARTRIDGE 7.62MM, M527

1. SCOPE

1.1 Scope.- This specification covers one type of ammunition container intended for packing small caliber ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636 Boxes, Fiberboard
TT-C-490 Cleaning methods and Pre-treatment of Ferrous Surfaces for Organic Coating

STANDARDS

Federal Test Method Std. No. 151 - Metals; Test Methods

Military

MIL-STD-105 Sampling Procedures and tables for Inspection by Attributes
MIL-STD-109 Quality Assurance Terms and Definitions
MIL-STD-129 Marking for Shipment and Storage
MIL-STD-406 Visual Inspection Standards for Tern Plate Cans and Steel Boxes Used in Small Arms Packing
MIL-STD-414 Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

U.S. Army Munitions Command

F10534757 Shipping and Storage Container, Cartridge 7.62mm: M527
IEL-10534757 List of Inspection Equipment for Box, Shipping & Storage Container, M527.

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, components and subassemblies shall comply with the applicable drawings, referenced specifications, and the following:

3.2 Material.- The contractor shall be responsible for selecting from the classification list, contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.2.1 Physical and Chemical Properties.- The physical and chemical properties specified on the applicable drawings shall be uniform throughout each part within specified tolerances.

3.3 First Article Sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Manufacturing process.- Components and assemblies shall be manufactured by a process approved by the contracting officer, and no deviations from that process shall be made without prior approval.

3.5 Assembly.- Components and assemblies for Containers M527 shall be combined to produce an assembly which conforms to the requirements of Drawing F10534757.

3.6 Protective finish.- When protective finish is required by component or assembly details, finish requirements shall be as required on the drawing and referenced specification and shall apply to all surfaces of the assembled container regardless of contour configuration.

3.6.1 Adhesion.- The protective finish shall not peel, flake, or fail to adhere after a minimum of 24 hours of drying.

3.6.2 Salt spray.- The finished container shall withstand 120 hours exposure to a salt spray of a five percent sodium chloride solution.

3.7 Functioning.

3.7.1 Cover Assembly.- The cover of the container shall open, close and be removed without binding or requiring exertion of undue force. The gasket shall not stick to top edges of the container, be cut or split, nor shift within the gasket retainer when the cover is removed.

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3.7.2 Hasp and latches.- Mating parts of the body hasps and latches shall engage without requiring deformation of any container part, and the latches shall close and open freely. When closed, the latches of the assembled container shall remain closed until manually opened.

3.7.3 Elevated temperature storage.- Following storage of the container in its closed position at an air temperature of 165 plus or minus 2 degrees Fahrenheit (F) for a period of 24 hours and subsequent return to ambient temperature, the cover shall function as required in 3.7.1.

3.8 Airtightness.- The assembled container shall not release more than 4 bubbles of air when subjected to an air pressure differential of 3 pounds per square inch (PSI) for 15 seconds.

3.9 Welds and hardware components.

3.9.1 Latch assemblies.- The latch assemblies shall withstand a pull of 600 pounds for a period of one minute without breakage or distortion of any of the components or welds.

3.9.2 Hasp assemblies.- The assembly of the hasps to the container shall withstand a pull of 900 pounds for one minute without failure of the hasps or weld.

3.9.3 End handle assemblies.- The end handle assemblies shall withstand a pull of 300 pounds for one minute without breakage or distortion of any of the components or welds.

3.10 Interchangeability (Cover and body assembly).- Fabrication of body assemblies and cover assemblies shall be such that these assemblies can be interchanged at random to produce containers which comply with the requirements of 3.7.1, 3.7.2, 3.8.

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects.- The containers shall be free of burrs, rough spots, sharp projection, cracks, splits, buckles or sintered surfaces.

3.11.2 Foreign matter.- The container shall be free of corrosion, chips, dirt, grease and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the con-

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tract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Ninety-six (96) assembled painted containers.
b. Five (5) sets of unassembled and unpainted parts.

The sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications, at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to meet the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot

4.3.1.1 Submittal of product.- The contractor shall submit the product in accordance with Standard MIL-STD-105.

4.3.1.2 Size of lot.- The lot size shall be as agreed upon between the contractor and the contracting officer provided the formation of the lot is in accordance with Standard MIL-STD-105.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis, using the applicable sampling plans and acceptance criteria of Standard MIL-STD-105 for the Acceptable Quality Levels (AQL's) pre-

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scribed in the classification of defects. All non-conforming containers and packed shipping boxes shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects and applicable AQL's shall be as follows:

4.3.2.1.1 Box, Shipping and Storage Container, Cartridge, 7.62 Millimeter: M527 (See Drawing F10534757) 1/

Category, AQL and defects	Method of Inspection
Critical: None defined	
Major (AQL 1.00%)	
101. Incomplete manufacture	Visual
102. Incorrect assembly	Visual
103. Burrs, rough spots, sharp projections, cracks, splits, buckles	Visual
104. Incomplete or inadequate paint coverage	Visual
105. Length, outside, max	Gage
106. Width, outside, max	Gage
107. Height, outside, max	Gage
108. Length, body assembly inside, min	Gage
109. Width, body assembly inside, min	Gage
110. Height, body assembly inside, min	Gage
111. Concavity or convexity of cover assembly	Gage
112. Concavity or convexity of body assembly	Gage
Minor (AQL 2.50%)	
201. Chips, dirt, grease, other foreign matter, and other defects	Visual
202. Surface imperfections, excessive paint, paint not dry	Visual
203. Manufacturer's identification missing, or incomplete	Visual

4.3.2.1.2 Box, Shipping for: Shipping and Storage Container, Cartridge, 7.62 Millimeter; M527

Category, AQL and defects	Method of Inspection
Critical: None defined	
Major: None defined	

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Category, AQL and defects	Method of Inspection
Minor (AQL 2.50%)	
201. Contents loose	Manual/visual
202. Contents incorrect	Visual
203. Separator missing	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Box inadequately or improperly sealed	Visual
207. Markings missing, incorrect or illegible	Visual

1/ In the examination for visual defects MIL-STD-406 shall be used as a basis of comparison for acceptance, where applicable, except that all defects classified as "Incidental" in MIL-STD-406 shall be considered permissible.

4.3.3 Tests.- The following tests shall be performed on each lot in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Failure on one or more of these tests shall be without penalty to the lot, but the manufacturer shall be required immediately to correct his surface preparation and/or painting practice, as pertinint, to comply with applicable specification.

4.3.3.1.1 Adhesion.- A sample of 3 containers shall be subjected to the tape test. Test failure is defined as one or more test containers showing any bare metal or underlying phosphate coating on either the cover or the body assembly.

4.3.3.1.2 Salt spray.- A sample of 3 containers shall be subjected to the salt spray test. Test failure is defined as one or more test containers showing rust spots, wrinkles, or blisters on any of the body or cover assembly surfaces, or evidence of rust creepage extending more than 1/8 inch on either side of the scribe marks, or rusting of part of the hardware subject to pressure or abrasion in normal use, or rust exceeding the permissable standards of MIL-STD-406, or enamel softer than unexposed enamel after air drying for 24 hours following the test.

4.3.3.1.3 Film thickness.- A sample of 3 containers shall be checked for film thickness. Test failure is defined as one or more test containers not conforming to the drawing requirement.

4.3.3.2 Functioning

4.3.3.2.1 Cover assembly.- Each container of the examination sample (4.3.2.1.1) shall be tested for compliance with the requirement of 3.7.1. Failure of the sample to meet acceptance criteria of MIL-STD-105 for an AQL of 1.0% shall be cause for rejection of the lot.

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4.3.3.2.2 Hasps and Latches-Each container of the examination sample (4.3.2.1.1) shall be tested for compliance with the requirement of 3.7.2. Failure of the sample to meet the acceptance criteria of MIL-STD-105 for an AQL of 1.0% shall be cause for rejection of the lot.

4.3.3.2.3 Elevated temperature storage.- A sample of 35 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for compliance with 3.7.3. Failure of any container to comply with the requirement shall be cause for rejection of the lot.

4.3.3.3 Airtightness.- A sample of 35 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for compliance with 3.8. Failure of any container to comply with the requirement shall be cause for rejection of the lot.

4.3.3.4 Welds and hardware components.- The same 35 containers used for the airtightness test (4.3.3.3) shall be tested for compliance with the requirements of 3.9. Failure of any component to meet the applicable requirements shall be cause for rejection of the lot.

4.3.3.4.1 Disposition of tested boxes.- Those containers which have successfully passed the tests prescribed in 4.3.3.4 may be returned to the lot provided they are subsequently tested for airtightness (4.3.3.3) and ambient temperature functioning (4.3.3.2.1 and 4.3.3.2.2) and are found acceptable.

4.3.3.5 Gasket compression test.- A sample of 10 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for Gasket Compression. The variables sampling plan of MIL-STD-414 example C3 for an AQL of 1.00 percent shall be used. Failure of any container to comply with the drawing requirements of failure of the sample to meet the criteria of the variables sampling plan, shall be cause for rejection of the lot.

4.3.3.6 Interchangeability (cover and body assembly).- A sample of 5 acceptable containers shall be selected from the examination sample (4.3.2.1.1) and tested for compliance with the requirements of 3.10. Failure of more than one assembled container to comply with the requirements of 3.7.1 and 3.7.2 or failure of more than one assembled container to comply with the requirement of 3.8 shall be cause for rejection of the lot.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10534757.

4.4 Test methods and procedures

4.4.1 Protective finish

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4.4.1.1 Adhesion.- This test shall be conducted in accordance with the paragraph of TT-C-490 applicable to the test for paint adhesion. Three places, randomly located, on the surfaces of each sample container body and cover assembly shall be tested to assure adequate paint adhesion on all surfaces.

4.4.1.2 Salt spray.- The sample shall be tested as specified in Federal Test Method Std No. 151, Method 811, for a period of 120 hours with a five percent solution of sodium chloride. Samples shall be so positioned that the one edge of each assembled container is elevated approximately 15 degrees from the horizontal and rests upon a wooden support while the cover assembly face is parallel to the principal direction of horizontal flow of the fog through the chamber. Prior to exposure, cross marks (x) approximately three inches long and penetrating to base metal shall be made with a machinists scriber (or equivalent) across the flat exterior portion of each side, the bottom, the hinge end, and the cover.

4.4.1.3 Film thickness.- Measurements shall be made at a random location on the exterior of each body assembly surface, a cover assembly surface and an accessible hardware surface. The test shall be conducted in accordance with paragraphs of TT-C-490 applicable to the test for paint thickness.

4.4.2 Functioning.- The sample containers, while at ambient temperature, shall be manually and visually examined to assure compliance with the requirements of 3.7.1 and 3.7.2.

4.4.3 Elevated temperature storage.- The sample containers shall be conditioned as prescribed in 3.7.3 and tested in accordance with 4.4.2.

4.4.4 Airtightness.- The sample containers shall be tested by submerging in ambient temperature water, maintaining them for 15 seconds at an air pressure differential of 3 pounds per square inch in excess of the outside pressure using an approved vacuum testing machine.

4.4.5 Welds and hardware components

4.4.5.1 End handle assemblies.- The end handle assemblies of the sample shall be tested for security of attachment to the box body by clamping one end handle to a test fixture in a suitable device and slowly applying the force specified in 3.9.3 for a period of one minute to the other end handle. During this test the box body shall rest upon its bottom, the force shall be applied parallel to the bottom with bearing against the ferrule of each handle.

4.4.5.2 Latch assemblies.- The sample shall be tested for security of welds and attachment of component parts of the assemblies (comprising the latches, latch links, latch link retainers, and covers) by fixing one latch assembly and slowly applying the force specified in 3.9.1 to the other latch assembly for a period of one minute. The force shall be applied parallel to the face of the cover and shall

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bear against the surface of the latch link exposed on the underside of each latch.

4.4.5.3 Hasp assemblies.- The container shall be clamped in a suitable fixture, resting on its bottom and with cover assembly removed. The force specified in 3.9.2 shall be slowly applied to each hasp in a vertical direction, with bearing against the underside of the offset of the hasp over the entire surface available, and maintained for a period of one minute. The container body shall be supported, to prevent distortion or collapse, by means of a snug fitting wood filler block.

4.4.6 Gasket Compression Test.- The sample shall be measured for Gasket Compression in the following manner: The box shall be positioned in a suitable fixture so that it cannot move. Six points, one on each end 3 1/2 inches from either side of the box body and two on each side 4 inches from each end of the box body, shall be located on both the box body and cover. With the box fully closed a reading at each point with a standard height gage shall be recorded, then the cover shall be removed and the height to the hemmed edge of the body at the same points shall be recorded. Subtraction of the second reading from the first reading gives the dimension of the cover plus compressed gasket. This value when subtracted from the constant calculated from the thickness of the cover and uncompressed gasket or from direct measurement of the cover and uncompressed gasket gives the compression at that point. Care must be exercised in this direct measurement so that the surface of the measuring device does not rest in the groove of the gasket caused by compression set; a shim may be used to assure the correct reading. For each sample box, the average of the six compression measurements shall be computed. The average values thus obtained shall be used in the application of the procedures of Standard MIL-STD-414.

4.4.7 Interchangeability.- The cover assembly shall be removed from each of the completely assembled containers and be reassembled in turn on each of the different container bodies from which the cover assembly was removed. Each reassembled container shall be tested for Functioning (4.4.2) and Airtightness (4.4.4).

5. PREPARATION FOR DELIVERY

5.1 Packing.- Level C.- Unless otherwise specified by the contracting officer six assembled containers fully closed shall be packed in a type CF, class domestic style RSC, grade 200, corrugated fiberboard carton conforming to PPP-B-636. Carton shall be sealed in accordance with appendix to apecification PPP-B-636, for class domestic boxes.

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5.2 Marking.- Each carton shall be marked and contain the following information:

a. Cosignee
b. Manufacturer
c. Item nomenclature
d. Lot number
e. Contract

and other markings as required by MIL-STD-129.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements. (See 4.2)

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of containers presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Intended Use.- Shipping and storage container, Cartridge 7.62mm M527 is intended for use in packaging and shipping of 7.62mm Ammunition.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No:

8140-A041

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